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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 02/05/2008 |
| **PI Name:** | Jin, Deborah Ph.D. | | |
| **Project Title:** | A strongly interacting gas of Fermi atoms | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FUNDAMENTAL PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Laser cooled atomic physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | UNIVERSITY | **Phone:** | 303-735-3740 |
| **Organization Name:** | University of Colorado - Boulder | | |
| **PI Address 1:** | JILA and Department of Physics | | |
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| **PI Web Page:** | | | |
| **City:** | Boulder | **State:** | CO |
| **Zip Code:** | 80309-0440 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2002 Physical Sciences 02-OBPR-03 |
| **Start Date:** | 04/01/2004 | **End Date:** | 09/30/2006 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | Maleki, Lute | **Contact Phone:** | 818-354-3688 |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | GSN 4518 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Ultracold gases of atoms have enormous potential for improving precision measurements, atomic time standards, and atom interferometers. In addition these gases provide a unique system for exploring quantum behavior such as superfluidity. However, at the ultralow temperatures of interest gravity becomes a dominant force on the atoms and provides a limitation on interaction times and achievable temperatures. Ground-based investigations, such as the work proposed here, can elucidate these limitations and the potential benefit of a microgravity environment. We propose to continue investigations of an ultracold gas of fermionic atoms. Ultracold Fermi gases of atoms have only been created in the last five years, and there remain exciting new opportunities for the study of basic quantum phenomena. The proposed work here concentrates on exploring interaction effects in the trapped gas. We will exploit atomic scattering resonances to control interactions in our dilute Fermi gas and probe the possibility of Fermi superfluidity in a gas, analogous to superfluid 3He and to superconductivity. | | |

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| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | We performed ground-based investigations of an ultracold, strongly interacting gas of Fermi atoms. Studies of ultracold atoms gases have the potential to impact precision measurements, atomic time standards, and atom interferometers. In addition, these gases provide a unique opportunity for exploring quantum behavior such as superfluidity in a well-controlled model system. The research accomplished in this project focused on effects of strong atom-atom interactions an ultracold Fermi gas. These interactions were precisely controlled using an atomic scattering resonance and thus we could create and investigate a superfluid state analogous to superfluid 3He and to superconductivity.<br>We explored Fermi condensates, which were first reported in January 2005 by my group. These condensates of pairs of fermionic atoms in an ultracold atomic gas provide a unique model system for exploring the macroscopic quantum phenomena of superconductivity and superfluidity. Furthermore, because the fermionic condensate occurs for a remarkably high relative temperature compared to the Fermi temperature, it is hoped that studies of this new system will ultimately enhance understanding of the technologically very important phenomena of high temperature superconductivity. |
| **Task Progress:** | Since June 2004 when the NASA funding for this project started, we completed a number of quantitative studies of the fermionic condensates. These include measurements of excitation spectra, momentum distributions and kinetic energy, and potential energy of the ultracold, strongly interacting Fermi gas. Fermionic condensates provide the first real experimental access to the predicted BCS-BEC crossover. Therefore an important goal of our experiments is to provide quantitative studies that can be compared to various theories describing the connection between Bose-Einstein condensation (BEC) and the Bardeen-Cooper-Schrieffer theory of superconductivity (BCS). To this end we have had collaborative efforts with theorists to compare measurements and predictions, and we've published three papers that are co-authored by theorists.<br>Another result of this project was to demonstrate a powerful new probe of quantum gases. This probing technique was proposed theoretically as a way to detect Cooper pairs of atoms in a fermionic condensate. The basic idea is to use noise fluctuations in absorption images to probe correlations in the atomic gas. For example, in the fermionic condensate one has Cooper pairs of atoms with equal and opposite momentum. This means that if you happen to find a larger than average number of atoms with one particular momentum then there must also be a larger than average number of atoms with the equal but opposite momentum. We have demonstrated this technique by using the dissociation of weakly bound molecules (dimers consisting of two fermionic atoms in a bound state) to create atom pair correlations. The pair correlations were then detected by analyzing noise fluctuations in absorption images. We could detect spatial correlations immediately after dissociation of the molecules, and further we could detect momentum correlations (such as described for Cooper pairs above) after allowing the atom gas to expand following the dissociation of the molecules.<br><br>The most recent on this project was the creation and study of p-wave Feshbach molecules. This work will impact future attempts to create non-s-wave superfluidity in an ultracold atom gas.<br><br>In conclusion, this project resulted in 9 publications, 5 of which were published in Physical Review Letters. Three graduate students and one postdoc worked on the project. One of the graduate students (Cindy Regal) completed a PhD and was awarded the American Physical Society's 2007 award for the best doctoral dissertation in Atomic, Molecular, and Optical physics. |
| **Bibliography Type:** | Description: (Last Updated: 02/06/2008) |
| **Abstracts for Journals and Proceedings** | Greiner M, Regal CA, Jin DS. "Fermionic condensates." XIX International Conference on Atomic Physics, ICAP 2004, Rio de Janeiro (Brazil), 25-30 July 2004.<br>Atomic Physics 19: XIX International Conference on Atomic Physics, AIP Conference Proceedings, Vol. 770, p. 209-217. http://dx.doi.org/10.1063/1.1928855 , May-2005 |
| **Articles in Peer-reviewed Journals** | Greiner M, Regal CA, Stewart JT, Jin DS. "Probing pair-correlated Fermionic atoms through correlations in atom shot noise." Phys Rev Lett. 2005 Mar 25;94(11):110401. PMID: 15903831 , Mar-2005 |
| **Articles in Peer-reviewed Journals** | Stewart JT, Gaebler JP, Regal CA, Jin DS. "Potential energy of a 40K Fermi gas in the BCS-BEC crossover." Phys Rev Lett. 2006 Dec 1;97(22):220406. PMID: 17155785 , Dec-2006 |
| **Articles in Peer-reviewed Journals** | Chen QJ, Regal CA, Greiner M, Jin DS, Levin K. "Understanding the superfluid phase diagram in trapped Fermi gasses." Physical Review A 2006 Apr;73(4):041601. http://dx.doi.org/10.1103/PhysRevA.73.041601 , Apr-2006 |
| **Articles in Peer-reviewed Journals** | Regal CA, Greiner M, Giorgini S, Holland M, Jin DS. "Momentum distribution of a Fermi gas of atoms in the BCS-BEC crossover." Phys Rev Lett. 2005 Dec 16;95(25):250404. PMID: 16384438 , Dec-2005 |
| **Articles in Peer-reviewed Journals** | Chen Q, Regal CA, Jin DS, Levin K. "Finite-temperature momentum distribution of a trapped Fermi gas." Physical Review A (Rapid Commun.), 2006 Jul;74(1):011601. http://dx.doi.org/10.1103/PhysRevA.74.011601 , Jul-2006 |
| **Articles in Peer-reviewed Journals** | Gaebler JP, Stewart JT, Bohn JL, Jin DS. "p-Wave Feshbach molecules." Phys Rev Lett. 2007 May 18;98(20):200403. PMID: 17677676 , May-2007 |
| **Awards** | Regal CA. "American Physical Society's 2007 award for Outstanding Doctoral Thesis Research in Atomic, Molecular, and Optical Physics, January 2007." Jan-2007 |
| **Books/Book Chapters** | Regal CA, Jin DS. "Experimental realization of the BCS-BEC crossover with a Fermi gas of atoms." in "Advances in Atomic, Molecular, and Optical Physics, vol. 54." Ed. P.R. Berman, C.C. Lin, E. Arimondo. Advances in Atomic, Molecular, and Optical Physics, vol. 54, p. 1-79. Academic Press, 2006. ISBN-10: 0-12-003854-4, Nov-2006 |

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| **Books/Book Chapters** | Jin DS, Regal CA. "Fermi gas experiments." in "International School of Physics Enrico Fermi, course CLXIV." Amsterdam : IOS Press, 2008. not yet released, Jan-2008 |
| **Dissertations and Theses** | Regal CA. "Experimental realization of the BCS-BEC crossover with a Fermi gas of atoms." Ph.D. dissertation, University of Colorado - Boulder, 2006. , Jun-2006 |